

Neuroanatomical and behavioral asymmetry in an adult compensated dyslexic [☆]

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Accepted 20 April 2006

Available online 5 June 2006

Abstract

Individual differences in cortical anatomy are readily observable, but their functional significance for behaviors such as reading is not well understood. Here, we report a case of an apparent compensated dyslexic who had attained high achievement in visuospatial mathematics. Data from a detailed background interview, psychometric testing, divided visual field tasks measuring basic word recognition (word naming, nonword naming, and lexical decision), and more controlled word retrieval (verb, category, and rhyme generation), and measurements of his atypical brain structure are described. The findings suggested that enhanced “top-down” processing could provide the means to compensate for deficient “bottom-up” word decoding skills in this case. Relative to controls, this individual also evidenced unusually large asymmetries on several divided visual field lexical tasks, an extreme leftward asymmetry of the planum temporale, and a rare form of Sylvian fissure morphology (Steinmetz type 4, [Steinmetz, H., Ebeling, U., Huang, Y., & Kahn, T. (1990). Sulcus topography of the parietal opercular region: An anatomic and MR study. *Brain and Language*, 38, 515–533.]). We suggest that certain forms of brain organization may be associated with successful behavioral compensation for dyslexia, and that anatomical variations in the right hemisphere may be important contributors to individual differences in reading acquisition and achievement.

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Keywords: Individual differences; Dyslexia; Planum temporale; Planum parietale; Cerebral asymmetries; Reading; Visual half-field; Sylvian fissure; Compensation

1. Introduction

Dyslexia is a developmental disorder that severely impacts the acquisition of reading skills, despite normal intelligence, and the absence of hard neurological signs. This specific reading disability is most often associated with deficient phonological processing (Gottardo, Stanovich, & Siegel, 1996;

Shankweiler & Lieberman, 1989; Wagner & Torgesen, 1987). Many dyslexic individuals are unable to achieve normal levels of reading comprehension, and remain dyslexic as adults (Felton, Naylor, & Wood, 1990; Flowers, 1995). However, others are eventually able to read fluently for comprehension and succeed in occupations that require high levels of literacy (Fink, 1998). Understanding the neural underpinnings of reading in such compensated dyslexics may allow us to identify alternate forms of brain organization that enable successful behavioral compensation.

Behaviorally, most studies indicate that difficulties with phonological processes persist in adults who have achieved good reading despite childhood diagnoses of dyslexia, as evident in tasks such as nonword reading

[☆] This research was supported by NSF Grant SBR-9729009 and NIH Grant DC006957. We are grateful to T.F. for his cheerful participation in this research and Brian Manowitz for image processing and measurements.

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(Bruck, 1990; Flowers, 1995; Gallagher, Laxon, Armstrong, & Frith, 1996), phoneme awareness (Bruck, 1992; Flowers, 1995; Gallagher et al., 1996), and spelling (Bruck, 1990; Gallagher et al., 1996; Lefly & Pennington, 1991). Although these studies differ in how compensation is operationally defined, there is substantial evidence that phonological deficits persist into adulthood despite the good reading comprehension that some childhood dyslexics ultimately achieve.

Since it appears that dyslexic deficiencies in phonological awareness are never eliminated regardless of age or level of reading achievement (Bruck, 1992; Lefly & Pennington, 1991), it is important to consider alternate reading mechanisms that may enable compensation. Some research suggests that dyslexics do not make greater than normal use of orthographic information during phonological word processing (Bruck, 1992). However, interactive compensatory models of reading suggest that well developed higher level reading subskills (e.g., reliance on context and semantic knowledge) can compensate for weaker lower level subskills (e.g., sound-letter associations) and contribute to individual differences in reading achievement (Rumelhart, 1977; Stanovich, 1980). Supporting this view, college-age dyslexics, relative to controls, show greater effects of sentence context on word naming (Ben-Dror, Pollatsek, & Scarpati, 1991). Similarly, higher level processes (listening comprehension, general knowledge, and vocabulary) moderate the relationship between phonological performance (nonword reading and phonological awareness) and reading comprehension in adults with a childhood diagnosis of dyslexia (Ransby & Swanson, 2003). This suggests that phonological skills may be less important for adult, than for child, readers. Although neither of these studies attempted to identify compensated adult readers, the findings imply that top-down processes may contribute to improvements in reading comprehension that some dyslexics achieve.

Even less is known about the neural substrates of reading in compensated dyslexics. To our knowledge no study of explicitly defined compensated dyslexics has explored cerebral asymmetries using behavioral methods. The most relevant study may be that of Atchley, Story, and Buchanan (2001) who studied college students with a history of developmental language disorder (DLD) using a divided visual field semantic priming paradigm. Activation of the subordinate meaning of ambiguous words was obtained for LVF/RH trials for control participants, but not for the DLD students. This finding could imply a RH word meaning access deficit in higher functioning adults with developmental language dysfunction.

A few neuroimaging studies have compared brain activity profiles associated with compensation. For example, Ingvar et al. (2002) observed that, relative to controls, the compensated dyslexics had a smaller spatial extent of RH perisylvian activation, but increased levels of activation in right medial frontal and perisylvian areas during spoken word reading. During silent word reading, the dyslexics also showed

increased activation in the right temporoparietal area. In a letter rhyme decision task compensated dyslexics had fewer regions activated than controls, with markedly less activation in the insula, left premotor, and Wernicke's area (Paulesu et al., 1996). An investigation by Shaywitz et al. (2003) utilized fMRI to compare normal readers, compensated dyslexics, and persistent dyslexics. Relative to controls, both dyslexic groups had underactivation of superior temporal and occipitotemporal regions, and overactivation of right inferior frontal areas during a nonword rhyme decision task. However, only the compensated dyslexics activated additional regions including right superior frontal and midtemporal areas, and left anterior cingulate gyrus. Although the tasks, subject groups, and imaging techniques vary greatly in these studies, the results suggest that impaired readers who learn to compensate utilize a somewhat different network of brain regions during reading than do normal readers or those with persistent dyslexia.

One factor contributing to variation in brain activity profiles could be structural variation in the underlying neural substrate. Leonard, Voeller, Lombardino, Morris, and Hynd (1993) compared Sylvian fissure morphology in a heterogeneous group of dyslexic individuals (including some compensated adults) with that of their unimpaired relatives and controls. They found an exaggerated *leftward* planar asymmetry and a "parietal shift" in the right hemisphere of three compensated individuals. A parietal shift occurs when the planum temporale posterior to Heschl's gyrus is severely truncated or absent while the planum parietale is relatively enlarged making a steep ascent directly posterior to Heschl's sulcus (V formation, Witelson & Kigar, 1992; type 4, Steinmetz, Ebeling, Huang, & Kahn, 1990). (Fig. 1 depicts various Sylvian fissure configurations in schematic form). This unusual planum temporale truncation results in a relative enlargement of the parietal lobe at the expense of the temporal lobe, more commonly in the right than the left hemisphere (Witelson & Kigar, 1992). In the Leonard et al. (1993) study, two of the dyslexics, but none of the controls or the relatives of the dyslexics, had parietal shifts bilaterally. A third compensated individual had a unilateral right parietal shift characterized by a rare subtype fissure called type 4 (Steinmetz et al., 1990). Here, not only is the planum temporale missing, but the parietal operculum is also truncated because the planum parietale rises anterior to the postcentral sulcus. In both type 4 and type V the posterior parietal lobe is enlarged. All three of the compensated dyslexics with parietal shifts were highly successful in professions that depended more on visuospatial than linguistic skill.

A subsequent study of compensated college students replicated the finding of greater leftward planum temporale asymmetry in the dyslexic individuals, particularly in those who had achieved good reading comprehension despite severe phonological deficits (Leonard et al., 2001). In this study, however, dyslexics and controls did not differ in the degree of parietal shift. Given the heterogeneity of behavioral and anatomical presentation in different

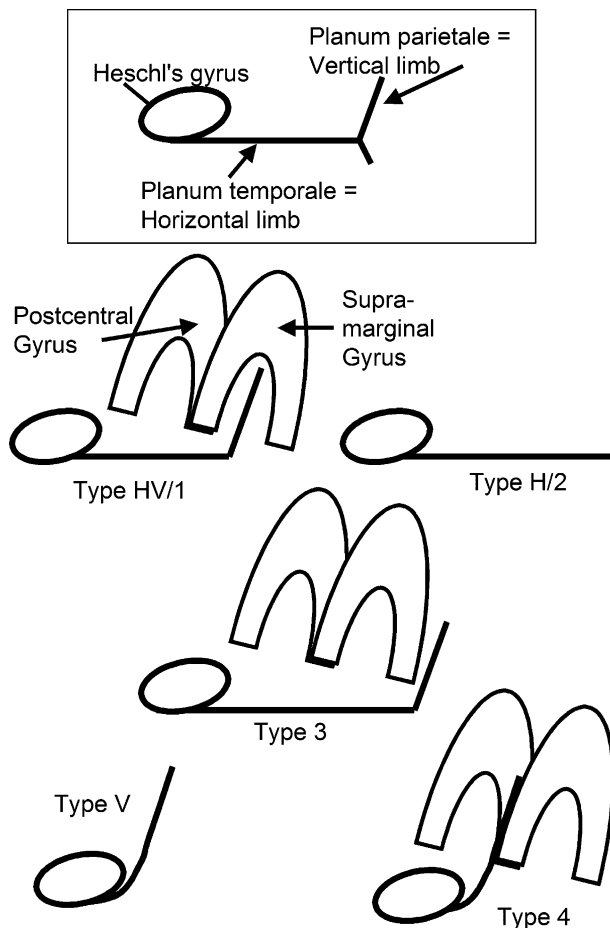


Fig. 1. Classification of Sylvian fissure types. Box: generic fissure. Witelson and Kigar (1992) described three fissure types based on the presence or absence of the planum temporale (horizontal limb) and planum parietale (vertical limb). Steinmetz et al. (1990) had previously proposed a similar classification based on the position of the planum parietale relative to the postcentral sulcus. In type 1 (Witelson HV), the planum parietale rises in the supramarginal gyrus. In type H/2 there is no planum parietale. In type 3, the planum parietale rises posterior to the supramarginal gyrus. In type V the planum temporale is absent because the planum parietale rises directly caudal to Heschl's gyrus. In type 4 the planum parietale merges or rises anterior to the postcentral sulcus. In both type 4 and type V, the planum temporale is small or absent. The two types differ in the relationship between the planum parietale and the postcentral sulcus. Type HV/1 is the most common in both hemispheres. Types H/2 and 3 are very rare on the right while type V/4 is very rare but may be seen in the right hemisphere.

dyslexic samples, additional attempts to associate anatomy and behavior in such individuals are clearly needed.

In this paper, we report a case study of a high-functioning individual whose constellation of abilities and deficits is characteristic of compensated dyslexia. We present information about his self-reported early history, his current performance on psychometric tests, functional laterality as indicated by a series of divided visual field word reading tasks, and some unusual neuroanatomical findings as measured from a structural MRI scan. Collectively, these data suggest an association between extreme structural asymmetries in posterior perisylvian regions and extreme behavioral asymmetries in tasks tapping low-level reading

functions that remained deficient despite superior reading comprehension.

2. Method

2.1. Participants

At the time of testing the participant (T.F.)¹ was a 28-year-old male graduate student in a social science field. He was a monolingual English speaker with normal visual acuity (20/25 on Snellen chart). He was strongly right handed as assessed by a 5-item hand preference questionnaire (score = +1.00) (Bryden, 1982), which yields an index ranging from +1.00 (extreme right handedness) to −1.00 (extreme left-handedness). His father is left-handed.

For each of the lateralized word recognition tasks, control data was available from 14 to 19 male students who had been tested in our laboratory for other studies (mean age = 21 years). All were native speakers of English, right handed (mean = +0.80), with normal or corrected-to-normal vision. Control data for the anatomical measurements came from 27 right-handed college students from an MRI archive at the University of Florida (mean age = 21.8 years).

2.2. Procedure

2.2.1. Sequence of assessments

As part of another study (Chiarello, Kacinik, Manowitz, Otto, & Leonard, 2004), the Word Identification, Word Attack, and Passage Comprehension subtests of the Woodcock Reading Mastery Test—Revised (WRMT-R, Woodcock, 1987) were administered in a single session. Following this, two test sessions were held in which several lateralized word recognition experiments were given (session 1—lexical decision, word naming; session 2—nonword naming, verb generation). The participant then received a structural MRI scan. Because the behavioral and neuroanatomical findings from this individual were unusual, we later contacted him to determine his willingness to participate in further behavioral testing and background interviews. He readily consented, and additional assessments were conducted over a period of 5 weeks, approximately 14 months after the original assessment. During this second testing period, consisting of 5 sessions, he was interviewed on several occasions and received two additional lateralized tasks (rhyme and category generation). T.F. was also given a series of psychometric tests that were selected to better characterize his cognitive abilities, to verify his self-reported deficits, and to probe functions that are frequently impaired in dyslexia. The tasks included the Raven's Advanced Progressive Matrices (Raven, Raven, & Court, 1998), spelling subtest of the Wide Range Achievement Test (WRAT 3, Jastak & Wilkinson, 1993), grammaticality

¹ T.F. is a pseudonym for our participant—these are not his true initials.

judgment and syntax construction subsets of the CASL (Carrow-Woolfolk, 1999), digits forward and backward of the WAIS-R (Wechsler, 1981), word comprehension subtests of the WRMT-R (Woodcock, 1987), math fluency and calculation subtests of the Woodcock-Johnson III Cognitive Abilities Assessment (WJ COG III, Woodcock, McGrew, & Mather, 2001), rapid digit and letter naming subtests of the Comprehensive Test of Phonological Processing (CTOPP, Wagner, Torgesen, & Rashotte, 1999), and sight word efficiency and phonemic decoding efficiency subtest of the Test of Word Reading Efficiency (TOWRE, Torgesen, Wagner, & Rashotte, 1999).

T.F. also completed the Adult Reading History Questionnaire (Lefly & Pennington, 2000). This measure was designed for adults to determine whether there is a childhood history of reading disability, and it has strong reliability and validity (Lefly & Pennington, 2000).

2.2.2. Divided visual field tests

Stimuli for all visual field experiments consisted of 3–6 letter concrete nouns, and pronounceable nonwords created by replacing a single letter of a concrete noun that was not part of the stimulus set. Letter replacements occurred equally often across all letter positions. All stimuli were presented in uppercase, black 20 pt Helvetica font on a white background screen, and subtended 1.7–2.8 degrees of horizontal, and 0.6 degrees of vertical, angle.

Power Macintosh 7500 computers with 17-inch Apple Vision monitors were used for stimulus presentation and recording of manual responses in the visual field tasks. Psycho programming software (Cohen, MacWhinney, Flatt, & Provost, 1993) was used to control experimental events and record responses. Participants were seated 60 cm in front of the monitor, using a headrest to stabilize head position. Participants used the ‘0’ and ‘.’ keys on the computer’s numeric keypad to register their lexical decision responses using the index and middle fingers of their right hand. A Sony ECM-MS907 microphone was used to register vocal responses. Accuracy of word and nonword pronunciation responses was coded online by the experimenter. A second monitor, located outside of the testing room, allowed the experimenter to view the stimulus presented on each trial, while listening to the participant’s response over headphones. Once a response was made, the experimenter entered an accuracy code into the computer. Special codes were entered for spurious vocal responses (a cough, for example), or failure to respond, and such trials were not analyzed. For the word generation tasks, the experimenter entered the participant’s response into the data file for later analysis.

Each experimental task began with a set of 30–46 practice trials, followed by the experimental trials (80 trials each for word and nonword naming, 160 for lexical decision, 66 for category generation, 88 for rhyme generation, and 92 for verb generation), with half of the items appearing in each visual field. Participants were instructed that the experiments investigated their ability to recognize words

they were not directly looking at. They were told to maintain fixation on the central ‘+’ fixation marker whenever it appeared on the screen, and to respond as quickly and accurately as possible. For word and nonword naming, participants were instructed to pronounce the stimulus, and in the lexical decision tasks to press response keys (labeled ‘word’ and ‘nonword’) to register their decisions. In verb generation, they were told to vocally produce a verb that went with each noun stimulus, that is, “what the object does, what it can be used for, or an action associated with it.” For rhyme generation, participants were asked to respond by saying a word that rhymed with the stimulus word, and for category generation, to produce the semantic category of the stimulus word.

On each trial the fixation marker appeared for 450–500 ms and flickered just prior to the onset of the stimulus. Then the stimulus word appeared randomly to the LVF or RVF, with the innermost edge of the stimulus 1.8 degrees eccentric from the fixation point. Stimulus target duration was determined based on previous research in our laboratory using these tasks (125 ms for word/nonword naming and lexical decision, 150 ms for the generation tasks). In all experiments, the next stimulus was presented two seconds after the participant’s response. A rest break was given at the mid-point of each experiment.

2.2.3. Magnetic resonance imaging (MRI) procedures

The volumetric MRI scan took place at the Computerized Diagnostic Imaging Center (CDIC), Riverside, CA in a GE 1.5T scanner equipped with Horizon echoplanar software. The field of view was 24 cm, and the voxel size was .94 × .94 × 1.2 mm. Images were concatenated into a series of sagittal images, rotated into the Talairach planes and reformatted into isotropic voxels using FSL (www.fmrib.ox.ac.uk) and programs written in PVWave (Visual Numerics, Boulder CO). After concatenation of the images into a volume, the brain was extracted from the surrounding tissues and automatically segmented into cerebrospinal fluid, gray, and white matter using FAST from the FSL toolkit. The output of this segmentation includes three volumes consisting of probabilistic partial volume estimates, one for each tissue class (Smith et al., 2004; Zhang, Brady, & Smith, 2001). BrainVisa (www.brainvisa.info) was used to create the surface renderings in Fig. 2. Following image preparation, several types of measurements were made as described below.

2.2.4. Quantitative manual measurements

Two raters blind to hemisphere as well as subject characteristics measured the surface areas of the planum temporale (PT), planum parietale (PP), and Heschl’s gyrus as described in Chiarello et al. (2004) and Leonard et al. (1993). Surface measurements of the pars triangularis were performed as described in Foundas, Eure, Luevano, and Weinberger (1998), and cross-sectional measurements of the corpus callosum were performed as described in Witelson (1989). At least two raters made each measurement and

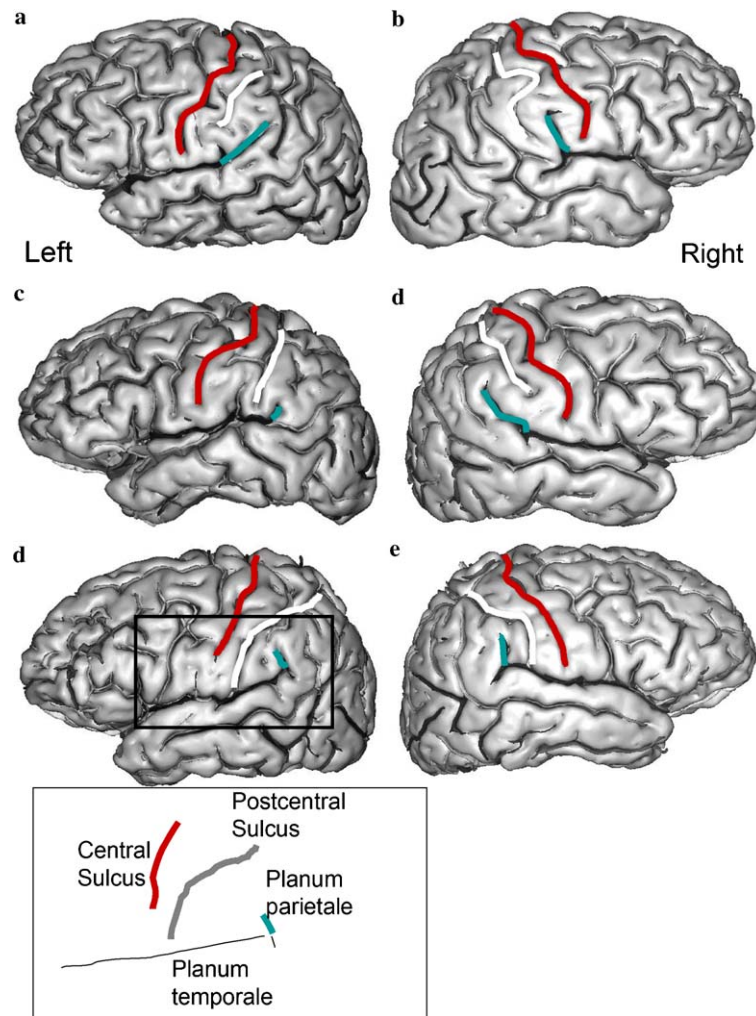


Fig. 2. Top: left and right hemisphere of participant T.F. (a) The left hemisphere has a typical type 1 formation with the central (red) and postcentral sulcus (white) descending anterior to the planum parietale (blue). (b) In the right hemisphere a type 4 fissure is seen: the planum parietale rises directly posterior to the central sulcus, anterior to the postcentral sulcus. (c–e) The left and right hemispheres of two participants from Chiarello et al. (2004) with typical type 1 formations and typical moderate behavioral asymmetries. The brain shown in (c) has a spurious posterior ascending ramus that arises anterior to the termination of the horizontal ramus and courses upwards into the postcentral gyrus. This spurious ramus is easily distinguished from a type 4 fissure because it originates laterally from a break in the upper bank of the horizontal ramus, rather than at its termination (Witelson & Kigar, 1992). Box: schematic of fissures.

differences were resolved by discussion. The coefficient of *interhemispheric* asymmetry for each structure was calculated by dividing the left/right difference by the left/right average. By convention, when the structure on the left is larger, the coefficient is positive. A quantitative index of the parietal shift is provided by the *intra*hemispheric asymmetry quotient for the planum temporale and planum parietale, calculated using a similar formula. When this quotient is negative, the planum parietale is larger than the planum temporale. When the planum temporale is larger, the coefficient is positive. Inter-rater reliabilities were calculated using an intraclass correlation and were all above 90% (Bartko & Carpenter, 1976).

2.2.5. Lobar measures of gray and white matter

The absence of consistent visible boundaries between the parietal, temporal, and occipital lobes on the lateral surface

of the brain makes it necessary to use arbitrary landmarks to separate the lobes. To avoid such arbitrary subdivisions, the decision was made to divide each hemisphere into two functionally relevant subdivisions at the central sulcus. This boundary is consistent with physiological data supporting a division between posterior regions whose major function is updating environmental information (parietal, temporal, and occipital lobes) and response selection regions in the frontal lobe (Passingham, 1993). The central sulcus was manually outlined on four selected sagittal images in each hemisphere of each brain. The images were chosen at sagittal positions where preliminary observations had suggested that the slope of the sulcus changed enough to significantly affect lobar volume. The manual outline (which excluded insula, internal capsule, and basal ganglia) was then automatically projected on a movie of each intervening image to ensure that it roughly coincided with the course of the

sulcus. To obtain grey and white matter volumes in the frontal and posterior regions, the voxels in frontal lobe anterior to the central sulcus and voxels in the parietal, temporal, and occipital lobes posterior to the central sulcus were summed and multiplied by the mean partial volume estimate for the grey and white tissue classes in that region as described in the FSL documentation.

2.2.6. Qualitative Sylvian fissure classification

The Sylvian fissure and parietal operculum were classified using two alternate methods (see Fig. 1). The Steinmetz classification (types 1–4) distinguishes both the presence and absence of the planum parietale and planum temporale, as well as the location of the planum parietale in relation to the postcentral and supramarginal gyri (Steinmetz et al., 1990). The Witelson classification (HV, V, and H) does not use this second criterion, and considers only the presence or absence of the horizontal (planum temporale) and vertical (planum parietale) limbs. Types HV and 1, and H and 2 are equivalent. As shown in Fig. 1, in type HV/1, the planum parietale rises in the supramarginal gyrus. In type H/2 no planum parietale is present. In type 3, the planum parietale rises posterior to the supramarginal gyrus, and in type 4, the planum parietale merges with or rises anterior to the postcentral gyrus. Although most type 4's are type V, the reverse is not the case because in most type V fissures the planum parietale rises posterior to the postcentral sulcus in the supramarginal gyrus in the normal fashion. Type HV/1 is the most common in both hemispheres. Type H/2 is very rare on the right, while type V/4 is very rare, but when present is usually found on the right (Witelson & Kigar, 1992).

3. Results

3.1. Biographical interview data

T.F. came from a middle-class family. His mother is a homemaker (high school graduate) and his father is a college-educated accountant. T.F. noted that while his father is creative and enjoys writing poetry, he does not read much. T.F. never received a diagnosis of a learning disability, but he did have an unusual educational history. He reported that his kindergarten teacher suspected he was mentally retarded, but a psychological evaluation at that time was normal except for color blindness. He received speech therapy for a /w/ for /r/ substitution from 4th through 6th grade. T.F. recalled long-standing problems with letter reversals, and stated that he was unusually active as a child. He was a very poor student during elementary and secondary school and did not read for pleasure. He commented on difficulties performing simple calculations, and took the lowest possible level of math courses in high school. T.F. also found writing and grammar to be difficult. He failed English every semester in high school and attended summer school for remedial English every year. He frequently skipped school, and barely graduated high school.

T.F. enrolled in a community college, and was initially attracted to his field because he thought it involved little math. He became intrigued by his field, and transferred to a four-year institution. He stated that he “taught himself” study skills, trigonometry, and geometry, because he had never learned them in school. He earned straight As in his major, and then entered graduate school. His scores on the Graduate Record Examination, which were taken 4 years before his participation in our study, were 440 (verbal), 690 (quantitative), and 750 (analytical).

T.F. excelled in graduate school, and conducted research on two-dimensional (geometrical) modeling of human data. He also taught statistics successfully, although he reported embarrassment about his poor ability to perform simple calculations during class demonstrations. While in graduate school he had several first-authored publications, and completed his Ph.D. work in the year following our data collection. T.F. currently has a tenure-track Assistant Professor position at a well-known university.

T.F.'s self-reports of his cognitive strengths and weaknesses are also interesting. Although he describes himself as being very mathematical, he stated that he does not know “all the rules of math” and that he “needs to relate numbers to reality.” He was never able to memorize the times tables above 7, and learned how to multiply by 9 using a trick [e.g., 9×3 is computed by subtracting 1 from the second digit ($3 - 1 = 2$) to determine the first digit of the answer, then computing x from $2 + x = 9$, to determine the second digit of the answer]. However, he excels at higher, more abstract levels of math, particularly if they involve visualization (e.g., T.F. states he is able to visualize trigonometric functions, and that he sees the relationships rather than the numbers). He believes that data analyses are his forte. T.F. commented that he learned how to write while in graduate school. He typically proofs his letters and e-mail many times before sending, and will do 6–7 drafts of a paper before showing it to anyone. T.F. believes that he has become quite good at the organizational aspects of writing scientific papers, but still has difficulty with spelling and punctuation.

In the areas of memory and language retrieval, T.F. reports that he experiences difficulties that he attempts to compensate for on a regular basis. He reports difficulty remembering sequences of numbers, and noted that it took him 6 months to remember his home phone number and that he continues to have difficulty remembering his ATM pin number (recalling the correct digits in the wrong sequence). He has great difficulty remembering the names of his students and others that he is introduced to. In spite of his concentrated efforts to hold names in memory, he reports that names typically leave his memory immediately. During speech production, when he tries to retrieve a word, he states that he will often create a similar but incorrect word such as “agnonymous” for anonymous. He reports having the greatest difficulty remembering the precise pronunciation of multisyllabic words. During spontaneous conversation, T.F. reports using fillers such as “um” while

searching for the right word or phrase. He feels that his unrehearsed speech is often dysfluent but that he does not stutter. However, casual conversations with T.F. do not reveal any deficits in speech production.

3.2. Psychometric test data

The results of the standardized tests, and percentile scores, are reported in Table 1. An estimate of T.F.'s non-verbal intelligence on the Raven's Matrices indicated that T.F. is an intelligent person whose word-level skills are well below the expected levels of performance for his academic standing.

On the reading assessments, T.F. excelled in passage comprehension. However, his performance on subtests involving word and nonword reading aloud was extremely low (13–35%), given his level of educational attainment and very good passage comprehension. His word comprehension is above average but not as high as would be expected given his superior reading comprehension. This suggests that he may be using contextual and semantic knowledge to process written text rather than advanced vocabulary knowledge that results from frequent reading. The examiner also noted numerous mispronunciations and omissions on the word comprehension test. Overall, the reading assessments document strength in text comprehension despite very poor word decoding.

The tests also revealed difficulty with spelling and grammar. T.F. was unable to correctly spell words such as *opportunity* and *possession*. The examiner also noted slow responses to many items in the syntax construction subtest. T.F.'s rapid naming skills were average and comparable to his word decoding and identification skills rather than his reading comprehension. Similarly, T.F. exhibited an average score for repeating digits forward. However, when his verbal working memory was taxed by repeating digits back-

wards, his score dropped below one standard deviation from the mean.

T.F.'s performance on the math subtests also indicate difficulty with calculation, particularly for the WJ COG III Math fluency which consists of very simple problems (e.g., $4 \times 4 = ?$ T.F. wrote 32) performed within a 3-min time limit. For the subtest that consisted of both simple mathematical operations as well as geometric, trigonometric, and calculus operations (WJCOG III Calculation), he achieved a much better score. However, for a Ph.D. whose research involves mathematical modeling, T.F.'s math scores are quite low.

On the Adult Reading History Questionnaire (Lefly & Pennington, 2000) T.F.'s score was 0.69. Scores that exceed 0.40 are considered diagnostic of a positive history of childhood reading disability, even for individuals who do not currently qualify for a diagnosis of dyslexia.

3.3. Divided visual field tests

Mean accuracy and RT for each task by visual field are given in Tables 2 and 3, respectively, for T.F. and the

Table 2

Accuracy data (percent correct) by visual field for T.F. and mean (standard deviation) of control subjects for divided visual field tasks

	LVF/RH		RVF/LH	
	T.F.	Control mean	T.F.	Control mean
Word naming	29.7*	82.0 (15.4)	90.0	90.4 (8.0)**
Nonword naming	22.5*	78.5 (16.3)	80.0	85.1 (11.4)**
Lexical decision	37.5*	78.4 (11.5)	95.0	85.5 (10.4)**
Verb generation	90.7	94.6 (6.4)	100	96.8 (3.2)
Category generation	100	98.5 (2.0)	100	98.7 (1.9)
Rhyme generation	91.2	93.9 (5.4)	97.4	95.6 (5.3)

* Accuracy for T.F. more than 2 standard deviations below the control mean.

** VF difference significant for control group, $p < .01$.

Table 1

Standardized test results for T.F.

Category	Skill (measure)	Score*	Percentile
Nonverbal IQ	(Ravens Advanced Progressive Matrices)	29	86
Reading	Untimed word reading aloud (WRMT-R Word Identification)	93	32
	Timed word reading aloud (TOWRE Sight Word Efficiency)	83	13
	Untimed nonword reading aloud (WRMT-R Word Attack)	92	29
	Timed nonword reading aloud (TOWRE Phonemic Decoding Efficiency)	94	35
	Word comprehension (WRMT-R Word Comprehension)	109	72
	Text comprehension (WRMT-R Passage Comprehension)	124	95
Spelling	Untimed written spelling of spoken words (WRAT3)	104	61
Grammar	Grammaticality judgment (CASL Grammaticality Judgment)	92	30
	Syntax Construction (CASL Syntax Construction)	100	50
Rapid naming	Letter naming (CTOPP Rapid Letter Naming)	10	50
	Digit naming (CTOPP Rapid Digit Naming)	10	50
Span memory	(WAIS-R Digits Forward)	11	52
	(WAIS-R Digits Backward)	6	14
Math	Timed arithmetic computations (WJ COG III Math Fluency)	96	39
	Untimed computations and solving equations (WJ COG III Calculation)	112	78

* Note: Scores listed are standard scores for all tests except WAIS digit spans, and Raven's Matrices.

Table 3

Reaction time data (ms) by visual field for T.F. and mean of control subjects for divided visual field tasks

	LVF/RH		RVF/LH	
	T.F.	Control mean	T.F.	Control mean
Word naming	1410*	826 (166)	737	728 (104)**
Nonword naming	1207*	859 (172)	853	824 (183)**
Lexical decision	919	790 (155)	865	752 (117)
Verb generation	1381	1578 (203)	1490	1563 (275)
Category generation	1266	1463 (282)	1336	1379 (249)**
Rhyme generation	2306	1896 (381)	2026	1806 (298)

* Reaction time for T.F. greater than two standard deviations from control mean.

** VF difference significant for control group, $p < .05$.

control group. The control subjects evidenced a RVF/LH advantage in these tasks.

There are two striking aspects of T.F.'s divided visual field performance, relative to the control group. First, T.F. performed quite well in all of the word generation tasks, for both LVF and RVF trials, although T.F.'s latencies for rhyme generation were somewhat slower than the control mean. In contrast, the word recognition tasks (word and nonword naming, lexical decision) revealed dramatic deficits for T.F.'s performance, but only for words presented to the LVF/RH. Examination of Tables 2 and 3 reveals that T.F.'s RVF/LH RT and accuracy performance was quite similar to the control group mean for word and nonword naming. Yet his LVF/RH scores were much slower and far less accurate (<30% correct) than controls (78–82% correct), falling more than 2 sd's from the control mean. The lexical decision accuracy data followed this same pattern.

Asymmetry scores were calculated for both T.F. and the control participants using the standard laterality indices: Accuracy $[(RVF - LVF)/(RVF + LVF)]$; RT $[(LVF - RVF)/$

(LVF + RVF)]. Positive scores indicate a RVF/LH advantage. These data are provided in Table 4, and confirm that T.F.'s asymmetries are generally within the control range for the word generation tasks. However, T.F. obtained very exaggerated asymmetries, favoring the RVF/LH, for the word and nonword naming tasks. His laterality scores were well outside the control range for these tasks. This was true as well as for the lexical decision accuracy asymmetry, although the RT asymmetry for this task was not remarkable.

In sum, on the visual field tasks T.F. obtained extremely poor LVF/RH performance only for lower-level word recognition/decoding tasks, and not for the more deliberative word generation tasks. This resulted in exaggerated LH asymmetries for the former tasks that exceeded the entire range of scores obtained from the control participants.

3.4. Neuroanatomical findings

3.4.1. Quantitative manual measurements

The means and standard deviations for perisylvian measures are presented in Table 5 (double asterisks indicate the measures where T.F. exceeds 2 standard deviations from the mean). As can be seen, his right planum temporale is unusually small and his left planum parietale is unusually large. This pattern was not found in any member of the comparison group. With respect to interhemispheric asymmetries, the degree of leftward planum temporale asymmetry evidenced by T.F. was outside the entire range of the comparison group. In contrast, the interhemispheric asymmetry for the planum parietale was normal. With respect to the intrahemispheric asymmetry, in both of T.F.'s hemispheres the planum parietale was larger than the planum temporale, providing evidence of bilateral parietal shifts. No member of the comparison group showed this asymmetry (PP > PT) bilaterally. The right pars triangularis was somewhat large (although still within the range of control values). Measurements for T.F.'s total corpus callosum area and the five subregions were not unusual (see Table 6).

3.4.2. Lobar grey and white matter volumes

The means and standard deviations for frontal and posterior gray and white matter are presented in Table 7 (upper). Similar to controls, T.F.'s posterior regions were relatively larger than his frontal regions, and his area measurements were not unusual. Table 7 (lower) also indicates

Table 4

Laterality indices for T.F. and range of control subject scores for divided visual field tasks for accuracy (percent correct) and reaction time (ms)

	Accuracy		Reaction time	
	T.F.	Control range	T.F.	Control range
Word naming	+0.50	−0.07...+0.34	+0.31	−0.03...+0.25
Nonword naming	+0.56	−0.05...+0.24	+0.17	−0.08...+0.08
Lexical decision	+0.43	−0.10...+0.25	+0.03	−0.10...+0.08
Verb generation	+0.05	−0.02...+0.12	−0.04	−0.07...+0.11
Category generation	0	−0.02...+0.02	−0.03	−0.01...+0.07
Rhyme generation	+0.03	−0.04...+0.08	+0.07	−0.09...+0.12

Table 5

Manual surface area measurements for Sylvian fissure structures for T.F. and mean (standard deviation) of control participants ($N = 27$)

	Planum temporale (cm ²)		Planum parietale (cm ²)		Interhemispheric asymmetry		Intrahemispheric asymmetry		Pars triangularis (cm ²)		Interhemispheric asymmetry
	RH	LH	RH	LH	Planum temporale	Planum parietale	RH	LH	RH	LH	
T.F.	0.17**	1.92*	2.90	2.87**	1.67**	−0.01	−1.78**	−0.40**	4.05*	3.28	−0.21
Controls	2.20 (1.01)	3.44 (0.77)	1.82 (0.84)	1.21 (0.79)	0.50 (0.52)	−0.36 (0.95)	0.16 (0.86)	0.99 (0.53)	2.80 (0.82)	2.89 (0.78)	0.04 (0.46)

* Indicates values between 1.5 and 2 SD from the group average.

** Indicates measurements greater than 2 SD from the group average.

Table 6

Manual cross sectional area measurements for corpus callosum (CC) for T.F. and mean (standard deviation) of control participants ($N = 27$)

	Total CC	Rost-rum	Genu	Body			Isthmus	Splenum
				Ant	Mid	Post		
T.F.	6.73	0.15	1.55	0.99	1.01	0.80	0.47	1.74
Controls	6.92 (0.95)	0.26 (.07)	1.54 (.32)	1.06 (0.18)	0.86 (0.16)	0.75 (0.13)	0.64 (0.12)	1.82 (0.25)

Table 7

Automated tissue segmentation values for gray and white matter for T.F. and mean (standard deviation) of control participants ($N = 27$)

	Gray matter (cc)				White matter (cc)			
	Frontal		Posterior		Frontal		Posterior	
	RH	LH	RH	LH	RH	LH	RH	LH
T.F.	122	109	203*	198	99	88	166	166
Controls	121 (13)	112 (9)	174 (17)	177 (15)	104 (11)	98 (10)	161 (15)	165 (18)
	Interhemispheric asymmetry				Frontal proportion of hemisphere			
	Gray matter		White matter		Gray matter		White matter	
	Frontal	Posterior	Frontal	Posterior	RH	LH	RH	LH
T.F.	−0.107	−0.027	−0.115	0.002	.375*	.356*	.373	.346
Controls	−0.071 (.092)	0.013 (.047)	−0.064 (.043)	0.023 (.038)	.409 (.022)	.389 (.018)	.392 (.018)	.372 (.019)

* Indicates values between 1.5 and 2 SD from the group average.

gray and white matter asymmetries, and the proportion of each hemisphere occupied by the frontal lobes. T.F.'s interhemispheric gray and white matter asymmetries were well within the control range. However, in case T.F., the proportion of gray matter in the frontal lobes was somewhat reduced relative to controls.

3.4.3. Qualitative Sylvian fissure classification

Case T.F. had a typical Sylvian fissure in the left hemisphere (type 1/HV) but a highly atypical formation (type 4/V) in the right hemisphere (see Fig. 2). No other type 4 fissures were seen in the control group and none was seen in another comparison group of 37 male controls recruited for experiments in Florida (C.M. Leonard, unpublished data).

In summary, several brain measures were unusual in case T.F. Most notably, there was an exaggerated leftward planum temporale asymmetry (due to a very small right PT), and unusually large plana parietale (relative to the planum temporale). In addition, the highly unusual type 4 sylvian fissure was seen within the right hemisphere. Taken as a whole, T.F.'s Sylvian fissures showed very atypical morphology.

4. Discussion

Psychometric testing of T.F. provides a portrait of impaired and intact cognitive skills that largely confirmed his self-reports. We argue below that his cognitive profile is consistent with what has been reported for compensated dyslexics. Next, we discuss T.F.'s behavioral laterality findings in the context of his cognitive profile and neuroanatomical findings. We consider the implications of these findings, while being mindful of the inherent limitations in generalizing from a single case study.

4.1. Cognitive profile

The phonological deficits that accompany dyslexia are well-known and such difficulties frequently persist into adulthood, even for individuals who have achieved normal reading comprehension (Bruck, 1992, 1998; Lefly & Pennington, 1991). Reading aloud of nonwords may be a particularly sensitive marker of residual phonological impairment (Felton et al., 1990; Flowers, 1995; Gallagher et al., 1996; Snowling, Nation, Moxham, Gallagher, & Frith, 1997). The reading profile of our case is consistent with such findings. His scores for reading aloud of words and nonwords were at or below the 35th percentile, despite superior text comprehension. His word comprehension was above average but not as high as might be expected given his text comprehension level, suggesting that he is compensating by relying on the contextual information provided in connected text. It is interesting to note that a large-scale regression study of adults who had childhood diagnoses of dyslexia concluded that higher level processes such as listening comprehension and general knowledge moderated the relationship between phonology and reading comprehension (Ransby & Swanson, 2003). Hence, although we do not have childhood reading data for T.F., his current reading profile is consistent with compensated dyslexia. This view is supported by his score on the Adult Reading History Questionnaire (Lefly & Pennington, 2000) that was strongly indicative of a history of childhood dyslexia.

Given his high level of academic achievement, T.F. also performed relatively poorly on tests of spelling, grammar, rapid naming, and verbal working memory. Subtle deficiencies in these spoken language skills are frequently

associated with dyslexia, and may be a consequence of phonological processing weaknesses (Bishop & Snowling, 2004). For example, deficits among dyslexics have been documented on tasks of verbal working memory (Kibby, Marks, Morgan, & Long, 2004; Snowling, 1991), word retrieval (Murphy, Pollatsek, & Well, 1988), naming speed (Vukovic, Wilson, & Nash, 2004; Wolf, Bally, & Morris, 1986), production of complex words (Catts, 1989; Paulesu et al., 1996; Snowling, 1981), the comprehension and construction of complex syntactic forms (McArthur, Hogben, Edwards, Heath, & Mengler, 2000), and spelling (Gallagher et al., 1996; Hatcher, Snowling, & Griffiths, 2002). Thus, T.F.'s constellation of language and memory difficulties represents common associations of dyslexia.

Particular mention should be made of T.F.'s mathematical abilities and deficits. On the one hand, his ability to rapidly perform simple calculations was well below average, confirming his self-reports. Calculation difficulties frequently co-occur with developmental dyslexia (Fletcher, 2005; Malmer, 2000). On the other hand, his performance was significantly better on tests including higher level mathematical computations, and T.F. claims to visualize complex mathematic concepts in spatial terms. The absence of measures to quantify this individual's high level spatial mathematical abilities is a limitation of our study. However, real world evidence of T.F.'s superior achievement on tasks requiring high level mathematical abilities is clear from his research involving mathematical modeling, his refereed publications in this area, and his teaching of statistics and quantitative methods at the University level. In some respects, T.F.'s mathematical abilities parallel his reading skills. In both cases, poor low level skills (word decoding, arithmetic calculations) co-exist with superior higher level abilities (text comprehension, spatial mathematical reasoning). We consider further below the extent to which enhanced "top-down" processing could provide the means to compensate for T.F.'s deficient "bottom-up" processing.

4.2. Behavioral lateralization and anatomical findings

The divided visual field tasks administered to T.F. were selected to tap both basic word recognition and decoding processes (word and nonword naming, lexical decision), and more deliberative word retrieval and production (verb, category, and rhyme generation). It is interesting that T.F.'s performance cleaved along this processing dimension as well. Perhaps the most remarkable aspect of his performance was the extremely low LVF accuracy for the three recognition/decoding tasks despite normal accuracy for RVF trials. This resulted in very exaggerated asymmetries that were outside of the range we observed for the control subjects. The reaction time data were similarly discrepant, ruling out the possibility that speed-accuracy trade-offs could account for the unusual findings. Thus, T.F. performed normally when verbal material was presented to the left hemisphere, but was markedly defi-

cient when stimuli were presented to the right hemisphere. This could indicate an alteration in word processing and/or decoding within the right hemisphere. Alternatively, very poor LVF performance could indicate a partial inter-hemispheric disconnection, such that information presented to the right hemisphere could not be made available to left hemisphere speech and motor regions. However, there was no anatomical indication of hemispheric disconnection—T.F.'s corpus callosum measurements were very similar to those of the control subjects. Furthermore, word generation accuracy, which also requires access to left hemisphere speech production regions was well within the control range. Again, the reaction time data followed the same pattern, although T.F. was somewhat slower than controls on rhyme generation. Hence, T.F.'s visual field data may suggest some deficiency within the right hemisphere for word decoding and analysis. We note that Atchley et al. (2001) also obtained some evidence for altered right hemisphere language processing in their study of college students with histories of developmental language disorder.

T.F.'s accurate rhyme generation performance might seem surprising in light of the difficulties he demonstrated for word and nonword pronunciation. However, dyslexics frequently perform normally on rhyme tasks despite poor phonological awareness in tasks that require segmentation (Bruck, 1992; Snowling et al., 1997). Thus, T.F.'s accurate rhyme generation appears to be typical for adult dyslexic individuals.

These data suggest a clear dissociation between lower level word tasks that produced exaggerated left hemisphere advantages, and slower, more deliberative word retrieval tasks that were performed more normally. We earlier noted that T.F.'s reading and mathematical competencies indicated deficiencies in lower level functions despite superior achievement in high level functions that can utilize context and knowledge. Even T.F.'s self-reports of his current writing skills (i.e., good organization, poor spelling) reflect a division between top-down and bottom-up abilities. Hence, this dissociation is not restricted to lateralized tasks. Nevertheless in our lateralized measures poor performance on the simple word recognition tasks was only observed when stimuli were presented to the right hemisphere. It is very interesting, then, that the anatomical measurements also demonstrated RH structural anomalies.

For many of the anatomical measurements T.F.'s data could not be distinguished from the control subjects. For example, the interhemispheric asymmetries for gray and white matter measurements were similar to controls, as were the callosal measurements. However, there were three unusual features of T.F.'s neuroanatomy: a very small right planum temporale, a type 4 Sylvian fissure in the right hemisphere, and bilateral "parietal shifts" indicating enlarged parietal, relative to superior temporal, cortical regions. The small right PT resulted in a very extreme leftward asymmetry of the planum temporale. The co-occurrence in this individual of extreme left hemi-

sphere asymmetries in both behavioral and anatomical measures is intriguing, but is difficult to interpret with only a single case. However, there are some findings that suggest that this association may not be entirely coincidental. We described earlier reports of exaggerated left planum temporale asymmetries in some compensated dyslexics (Leonard et al., 1993, 2001). In a previous study of college student readers we noted an association between planum temporale and visual field asymmetries for lexical tasks (Chiarello et al., 2004). One recent study suggested that LVF/RH lexical decision scores correlated more strongly with a standardized reading measure than RVF/LH scores (Weems & Zaidel, 2004), which could imply that variations in right hemisphere processing contribute to individual differences in reading skill. Thus, there are previous reports of relationships between reading ability and LVF processing, visual field performance and planum asymmetries, and exaggerated leftward PT asymmetries in compensated dyslexics. The current case thus represents an example of all three associations within a single individual: compensated dyslexia, exaggerated planum asymmetry, and extreme RVF/LH advantages on simple word recognition tasks. Further study of additional dyslexic and normal readers will be needed to confirm these linkages, but we suggest that dyslexic profiles characterized by small RH PT structures and exaggerated behavioral asymmetries may be associated with the ability to achieve good reading comprehension despite persisting word decoding deficiencies. Such individuals may form a relatively small subgroup of dyslexics whose reading disabilities in childhood may go undiagnosed because of their superior comprehension skills.

In addition to the unusual asymmetries, T.F. had a rare type 4 right Sylvian fissure, and bilateral parietal shifts, indicating increased parietal, relative to superior temporal, cortex. No other type 4 fissures were seen in the control group reported here, nor in another comparison group of 37 male controls recruited for experiments in Florida. None was observed in the 55 children studied by Hiemenz and Hynd (2000). However, an unusually high proportion of type 4 fissures has been reported in several individuals from a family with a high incidence of wide PIQ/VIQ splits and dyslexia profiles (Craggs, Sanchez, Kibby, Gilger, & Hynd, *in press*). This formation has been seen previously in the right hemisphere of a boy with specific language impairment that resolved in adolescence, as well as a severely dyslexic man with an unexpectedly good professional outcome (Leonard et al., 1993), similar to T.F. However, in that study a type 4 fissure was found in the right hemisphere of one of the controls who showed no evidence of language problems (although a full language assessment was not performed). Although the extent to which Albert Einstein evidenced dyslexia or developmental language delay is unsettled (Brian, 1996; West, 1997), it is interesting that an analysis of Einstein's brain (Witelson, Kigar, & Harvey, 1999) concluded that the parietal operculum was absent, and the posterior parietal lobes were enlarged relative to the temporal lobes, due to the parietal planum rising caudal to the

central sulcus (i.e., type 4 fissure) bilaterally. In the Leonard et al. (1993) study, two dyslexic men had negative intrahemispheric asymmetry quotients (i.e., PP > PT) in both the left and right hemisphere. This anatomical oddity was particularly interesting because both individuals had attained the top rungs of their highly visuospatially dependent medical specialties, in spite of severe (but undiagnosed) difficulties with written language and phonological decoding.

Thus our findings from case T.F., along with previous data, imply some association between, on the one hand, atypical Sylvian fissure morphology and enhanced parietal development, and on the other hand, high functioning dyslexia and visuo-spatial talent. This apparent linkage is consistent with the differing functions imputed to the superior temporal (language) and posterior parietal (visuo-spatial) cortex (e.g., Chafee, Crowe, Averbeck, & Georgopoulos, 2005; Galletti & Fattori, 2002; Nakada, Fujii, Yoneoka, & Kwee, 2001). However, even if subsequent data confirm this association, causality would remain undetermined. Functional imaging studies would also be helpful in determining structure-function relations in future cases.

4.3. Avenues of compensation and talent

It remains a puzzle why some children with dyslexia eventually become competent readers, while others have lifelong struggles with literacy. In particular, it is unclear why some dyslexics can utilize top-down processing and context to successfully comprehend text, while others cannot. Equally intriguing is whether there is any association between dyslexia and visuo-spatial talent (Geschwind & Galaburda, 1985). Ultimately, the answers to such questions must come from large-scale studies of individual differences in cognition and brain organization. However, an in-depth case study approach, such as is employed here, can also shed light on these issues by illustrating patterns of deficit, compensation, and talent.

In her examination of 60 dyslexics who became successful professionals in fields requiring high levels of literacy, Fink (1998) noted that many, like our case T.F., were "late bloomers" who currently do considerable writing in their fields and contribute new knowledge in their areas of expertise. Fink (1998) notes passionate interest in a particular field as a motivator, and T.F.'s self-reports conform to this pattern. Although motivation is often overlooked as an explanatory construct in neuropsychological studies, it may be an important contributor to individual differences in compensation for dyslexia.

In addition, high intelligence may be one factor in successful compensation for dyslexia. T.F. is very intelligent, and was able to eventually achieve considerable academic success, despite weak school performance in elementary and secondary school. An association between IQ and compensation was noted in the fMRI study of Shaywitz et al. (2003). They observed significantly higher IQs in young adults identified as compensated readers than in those iden-

tified as persistently poor readers. Higher intelligence may facilitate the use of knowledge and context to compensate for poor word decoding. Interestingly, perisylvian and extrasyllabic regions have been implicated in the processing of language in context (Xu, Kemeny, Park, Frattali, & Braun, 2005).

We suggest that certain forms of brain organization may also enable successful behavioral compensation for dyslexia². On the basis of our study of T.F., and other data in the literature, we suggest that anatomical variations in the right hemisphere may be more important for *individual differences* in reading acquisition and achievement than left hemisphere structures. It is plausible to suggest that there is greater phenotypic variation in recruitment of the right hemisphere for language; left hemisphere involvement may be much more invariant, at least within the right-handed population. Structural variations may mediate these individual differences in function.

Similarly we have noted the co-occurrence of parietal shifts and visuospatial/mathematical talents in T.F. and other reported cases. We acknowledge that drawing conclusions from such associations on the basis of individual cases is controversial. For example, the Witelson et al. (1999) study of Einstein's brain was severely criticized in two accompanying commentaries on the grounds that the absence of the parietal operculum was not unusual, even in the left hemisphere (Galaburda, 1999) and no data were available on other people with like mathematical skills (Seitz, 1999). Galaburda concluded "the search for gross anatomical markers for greatness will go on, I am sure, but I suspect that it will continue to be as unproductive as it has been in the past" (Galaburda, 1999, p. 1821). In opposition to this interpretation, we think the scattered reports of unusual visuospatial ability and parietal enlargement suggest that the intense study of individual anatomy coupled with an in depth assessment of behavioral skills might provide a rich new paradigm for the investigation of structural/functional dependencies. Although individual brain formations may not reveal markers of "greatness", they may very well indicate markers of unusual combinations of cognitive strengths and weaknesses of the sort described by West (1997). The current case report represents one contribution to this approach.

References

- Atchley, R. A., Story, J., & Buchanan, L. (2001). Exploring the contribution of the cerebral hemispheres to language compensation deficits in adults with developmental language disorder. *Brain and Cognition*, 46, 16–20.
- Bartko, J., & Carpenter, W. (1976). On the methods of theory and reliability. *The Journal of Nervous and Mental Disease*, 163, 307–317.
- Ben-Dror, I., Pollatsek, A., & Scarpato, S. (1991). Word identification in isolation and in context by college dyslexic students. *Brain and Language*, 40, 471–490.
- Bishop, D. V. M., & Snowling, M. (2004). Developmental dyslexia and specific language impairment. *Psychological Bulletin*, 130, 858–886.
- Brian, D. (1996). *Einstein: A life*. New York: John Wiley & Sons.
- Bruck, M. (1990). Word recognition skills of adults with childhood histories of dyslexia. *Developmental Psychology*, 26, 439–454.
- Bruck, M. (1992). Persistence of dyslexics' phonological awareness deficits. *Developmental Psychology*, 28, 874–886.
- Bruck, M. (1998). Outcomes of adults with childhood histories of dyslexia. In C. Hulme & R. Malatesha Joshi (Eds.), *Reading and spelling: Development and disorders* (pp. 179–200). Mahwah, N.J.: Lawrence Erlbaum.
- Bryden, M. P. (1982). *Laterality: Functional asymmetry in the intact brain*. New York: Academic Press.
- Carrow-Woolfolk, E. (1999). *Comprehensive assessment of spoken language*. Circle Pine, MN: Academic Guidance Services.
- Catts, H. W. (1989). Speech production deficits in developmental dyslexia. *Journal of Speech and Hearing Disorders*, 54, 422–428.
- Chafee, M. V., Crowe, D. A., Averbeck, B. B., & Georgopoulos, A. P. (2005). Neural correlates of spatial judgement during object construction in parietal cortex. *Cerebral Cortex*, 15, 1393–1413.
- Chiarello, C., Kacinik, N., Manowitz, B., Otto, R., & Leonard, C. (2004). Cerebral asymmetries for language: Evidence for structural-behavioral correlations. *Neuropsychology*, 18, 219–231.
- Cohen, J., MacWhinney, B., Flatt, M., & Provost, J. (1993). PsyScope: An interactive graphic system for designing and controlling experiments in the psychology laboratory using Macintosh computers. *Behavior Research Methods, Instruments, & Computers*, 25, 257–271.
- Craggs, J.G., Sanchez, J.C., Kibby, M.Y., Gilger, J.W., & Hynd, G.W. (in press). Brain morphology and neuropsychological profiles in a family displaying dyslexia and superior nonverbal intelligence. *Cortex*.
- Felton, R. H., Naylor, C. E., & Wood, F. B. (1990). Neuropsychological profile of adult dyslexics. *Brain and Language*, 39, 485–497.
- Fink, R. P. (1998). Literacy development in successful men and women with dyslexia. *Annals of Dyslexia*, 48, 311–342.
- Fletcher, J. M. (2005). Predicting math outcomes: Reading predictors and comorbidity. *Journal of Learning Disabilities*, 38, 308–312.
- Flowers, D. L. (1995). Neuropsychological profiles of persistent reading disability and reading improvement. In C. K. Leong & R. M. Joshi (Eds.), *Developmental and acquired dyslexia* (pp. 61–77). Dordrecht: Kluwer Academic Publishers.
- Foundas, A. L., Eure, K. F., Luevano, L. F., & Weinberger, D. R. (1998). MRI asymmetries of Broca's Area: The pars triangularis and pars opercularis. *Brain and Language*, 64, 282–296.
- Galaburda, A. M. (1999). Albert Einstein's brain. *Lancet*, 354, 1821; author reply. *Lancet*, 354, 1822.
- Gallagher, A. M., Laxon, V., Armstrong, E., & Frith, U. (1996). Phonological difficulties in high-functioning dyslexics. *Reading and Writing*, 8, 499–509.
- Galletti, C., & Fattori, P. (2002). Posterior parietal networks encoding visual space. In H.-O. Karnath, D. Miller, & G. Vallar (Eds.), *The cognitive and neural bases of spatial neglect* (pp. 59–69). New York: Oxford University Press.
- Geschwind, N., & Galaburda, A. M. (1985). Cerebral lateralization: Biological mechanisms, associations, and pathology: I. A hypothesis and a program for research. *Archives of Neurology*, 42, 428–459.
- Gottardo, A., Stanovich, K., & Siegel, L. (1996). The relationships between phonological sensitivity, syntactic processing and verbal working memory in the reading performances of third-grade children. *Journal of Experimental Child Psychology*, 63, 563–582.
- Hatcher, J., Snowling, M. J., & Griffiths, Y. M. (2002). Cognitive assessment of dyslexic students in higher education. *British Journal of Educational Psychology*, 72, 119–133.

² We are not arguing that the source of T.F.'s compensation should be attributable to any single factor, either brain morphology or psychological phenomena such as intelligence and motivation. Rather it is likely that a complex interaction among many neural and nonneural factors conjointly affect the development of intelligence and language function and mediate individual variation in these abilities.

- Hiemenz, J. R., & Hynd, G. W. (2000). Sulcal/gyral pattern morphology of the perisylvian language region in developmental dyslexia. *Brain and Language*, 74, 113–133.
- Ingvar, M., af Trampe, P., Greitz, T., Eriksson, L., Stone-Elander, S., & von Euler, C. (2002). Residual differences in language processing in compensated dyslexics revealed in simple word reading tasks. *Brain and Language*, 83, 249–267.
- Jastak, J., & Wilkinson, G. S. (1993). *Wide range achievement test: Revision 3*. Wilmington, DE: Wide Range Publishers.
- Kibby, M. Y., Marks, W., Morgan, S., & Long, C. J. (2004). Specific impairment in developmental reading disabilities: A working memory approach. *Journal of Learning Disabilities*, 37, 349–363.
- Lefly, D. L., & Pennington, B. F. (1991). Spelling errors and reading fluency in compensated adult dyslexics. *Annals of Dyslexia*, 41, 143–162.
- Lefly, D. L., & Pennington, B. F. (2000). Reliability and validity of the adult reading history questionnaire. *Journal of Learning Disabilities*, 33, 286–296.
- Leonard, C. M., Eckert, M. A., Lombardino, L. J., Oakland, T., Kranzler, J., Mohr, C. M., King, W. M., & Freeman, A. (2001). Anatomical risk factors for phonological dyslexia. *Cerebral Cortex*, 11, 148–157.
- Leonard, C. M., Voeller, K. K. S., Lombardino, L. J., Morris, M. K., Hynd, G. W., et al. (1993). Anomalous cerebral structure in dyslexia revealed with magnetic resonance imaging. *Archives of Neurology*, 50, 461–469.
- Malmer, G. (2000). Mathematics and dyslexia—An overlooked connection. *Dyslexia*, 6, 223–230.
- McArthur, G. M., Hogben, J. H., Edwards, V. T., Heath, S. M., & Mengler, E. D. (2000). On the specifics of specific reading disability and specific language impairment. *Journal of Child Psychology and Psychiatry*, 41, 869–874.
- Murphy, L. A., Pollatsek, A., & Well, A. D. (1988). Developmental dyslexia and word retrieval deficits. *Brain and Language*, 35, 1–23.
- Nakada, T., Fujii, Y., Yoneoka, Y., & Kwee, I. L. (2001). Planum temporale: Where spoken and written language meet. *European Neurology*, 46, 121–125.
- Paulesu, E., Frith, U., Snowling, M., Gallagher, A., Morton, J., Fackowiak, R. S. J., & Frith, C. D. (1996). Is developmental dyslexia a disconnection syndrome? Evidence from PET scanning. *Brain*, 119, 143–157.
- Passingham, R. (1993). *The frontal lobes and voluntary behavior*. Oxford: Oxford University Press.
- Ransby, M. J., & Swanson, H. W. (2003). Reading comprehension skills of young adults with childhood diagnoses of dyslexia. *Journal of Learning Disabilities*, 36, 538–555.
- Raven, J., Raven, J. C., & Court, J. H. (1998). *Advanced progressive matrices*. Oxford: Oxford Psychologists Press.
- Rumelhart, D. E. (1977). Toward an interactive model of reading. In S. Dornic (Ed.), *Attention and performance VI*. Hillsdale, NJ: Lawrence Erlbaum.
- Seitz, J. A. (1999). Albert Einstein's brain. *Lancet*, 354, 1822–1823.
- Shankweiler, I. Y., & Lieberman, D. (Eds.). (1989). *Phonology and reading disability: solving the reading puzzle*. Ann Arbor: University of Michigan Press.
- Shaywitz, S. E., Shaywitz, B. A., Fulbright, R. K., Skudlarki, P., Mencl, W. E., Constable, R. T., et al. (2003). Neural systems for compensation and persistence: Young adult outcome of childhood reading disability. *Biological Psychiatry*, 54, 25–33.
- Smith, S. M., Jenkinson, M., Woolrich, M. W., Beckmann, C. F., Behrens, T. E., Johansen-Berg, H., et al. (2004). Advances in functional and structural MR image analysis and implementation as FSL. *Neuroimage*, 2(Suppl. 1), S208–S219.
- Snowling, M. (1981). Phonemic deficits in developmental dyslexia. *Psychological Research*, 43, 219–234.
- Snowling, M. J. (1991). Developmental reading disorders. *Journal of Child Psychology and Psychiatry*, 32, 49–77.
- Snowling, M., Nation, K., Moxham, P., Gallagher, A., & Frith, U. (1997). Phonological processing skills of dyslexic students in higher education: A preliminary report. *Journal of Research in Reading*, 20, 31–41.
- Stanovich, K. E. (1980). Toward an interactive-compensatory model of individual differences in the development of reading fluency. *Reading Research Quarterly*, 16, 32–71.
- Steinmetz, H., Ebeling, U., Huang, Y., & Kahn, T. (1990). Sulcus topography of the parietal opercular region: An anatomic and MR study. *Brain and Language*, 38, 515–533.
- Torgesen, J., Wagner, R., & Rashotte, C. (1999). *Test of word reading efficiency*.
- Vukovic, R. K., Wilson, A. M., & Nash, K. K. (2004). Naming speed deficits in adults with reading disabilities: A test of the double-deficit hypothesis. *Journal of Learning Disabilities*, 37, 440–450.
- Wagner, R., & Torgesen, J. (1987). The nature of phonological processing and its causal role in the acquisition of reading skills. *Psychological Bulletin*, 101, 192–212.
- Wagner, R. K., Torgesen, J. K., & Rashotte, C. A. (1999). *Comprehensive test of phonological processing*. Austin, TX: PRO-ED.
- Wechsler, D. (1981). *Wechsler adult intelligence scale—revised*. New York: The Psychological Corporation.
- Weems, S. A., & Zaidel, E. (2004). The relationship between reading ability and lateralized lexical decision. *Brain and Cognition*, 55, 507–515.
- West, T. G. (1997). *In the mind's eye*. Amherst, NY: Prometheus Books.
- Witelson, S. F. (1989). Hand and sex differences in the isthmus and genu of the human corpus callosum. A postmortem morphological study. *Brain*, 113, 799–835.
- Witelson, S. F., & Kigar, D. (1992). Sylvian fissure morphology and asymmetry in men and women: Bilateral differences in relation to handedness in men. *Journal of Comparative Neurology*, 323, 326–340.
- Witelson, S. F., Kigar, D., & Harvey, T. (1999). The exceptional brain of Albert Einstein. *Lancet*, 353, 2149–2153.
- Wolf, M., Bally, H., & Morris, R. (1986). Automaticity, retrieval processes, and reading: A longitudinal study in average and impaired readers. *Child Development*, 57, 988–1000.
- Woodcock, R. (1987). *Woodcock reading mastery tests, revised*. Circle Pines, MN: American Guidance Service.
- Woodcock, R., McGrew, K. S., & Mather, N. (2001). *Woodcock-Johnson III Tests of Cognitive Abilities*. Itasca, IL: Riverside Publishing.
- Xu, J., Kemeny, S., Park, G., Frattali, C., & Braun, A. (2005). Language in context: Emergent features of word, sentence, and narrative comprehension. *Neuroimage*, 25, 1002–1015.
- Zhang, Y., Brady, M., & Smith, S. (2001). Segmentation of brain MR images through a hidden Markov random field model and the expectation-maximization algorithm. *IEEE Transactions On Medical Imaging*, 20, 45–57.